Middle-Rung Law: physics and chemistry formulas written with the ether frequency v_e and the expansion rate H

Companion file to the OSF preregistration. Predictions fixed 2026-11-05. Author: John Lu Qianhe (约翰·路谦和). Normalization relation $4\pi^2r^3c^2/(T^2Ghv)=1$ after Zhang Xiangqian (2026, unpublished). All numbers use CODATA 2018 constants.

Middle-rung law. At every epoch t:
$\hbar \cdot G = c^5 / (4\pi^2 \cdot v_e(t) \cdot H(t))$, equivalently $v_e(t) \cdot H(t) = v_P^2$, with $v_P = \sqrt{(c^5/\hbar G)}/2\pi$ the Planck (human-boundary) frequency.
v_e and H both change with cosmic time; their product is fixed, so ħ and G stay constant.

1. Present values (H_0 = 61.8–63.5 km s^{-1} Mpc^{-1}; central 62.0)

Quantity	Central (H_0 = 62.0)	Range (61.8–63.5)
Expansion rate H (universe rung)	2.009×10^{-18} s^{-1}	2.003×10^{-18} – 2.058×10^{-18} s^{-1}
Ether frequency $v_e = v_P^2/H$	4.337×10^{102} Hz	4.235×10^{102} – 4.351×10^{102} Hz
Planck (human-boundary) frequency v_P	2.952×10^{42} Hz	constant
Invariant product $v_e \cdot H = v_P^2$	8.715×10^{84} Hz^2	constant
Rung ratio $N = v_P/H$	1.469×10^{60}	1.435×10^{60} – 1.474×10^{60}
Universe age (H_0t_0 = 0.876)	13.82 Gyr	13.49–13.86 Gyr
Mid-rung frequency $\sqrt{(H\ v_P)}$	2.435 THz	2.43–2.46 THz

Numerically: $\hbar \cdot G = 7.039\times10^{-45}$ $J \cdot m^3$ kg^{-1} $s^{-2} = c^5 / (4\pi^2 \cdot 4.337\times10^{102} \cdot 2.009\times10^{-18})$.

2. Physics formulas — symbolic form (v_e, H vary with time) and present-value form

Formula	Standard	Ether form (time-dependent $v_e(t)$, H(t))	Present-value form ($v_e \cdot H = 8.71\times10^{84}$ Hz^2)	Value
Mass–energy	$E = mc^2$	$E = mc^2$ (unchanged); knot frequency $f = mc^2/h$	$E = mc^2$	8.988×10^{16} J (m = 1 kg)
Planck relation	$E = h\nu$	$E = c^5\nu / (2\pi\ G\ v_e\ H)$	$E = c^5\nu / (2\pi\ G \cdot 4.34\times10^{102} \cdot 2.01\times10^{-18})$	6.626×10^{-19} J ($\nu = 10^{15}$ Hz)

de Broglie wavelength	$\lambda = h/p$	$\lambda = c^5 / (2\pi G v_e H p)$	$\lambda = c^5 / (2\pi G \cdot 8.71\times10^{84} \cdot p)$	7.274×10^{-10} m (electron, 10^6 m/s)
Newtonian gravity	$F = GMm/r^2$	$F = c^5 M m / (4\pi^2 \hbar v_e H r^2)$	$F = c^5 M m / (4\pi^2 \hbar \cdot 8.71\times10^{84} \cdot r^2)$	9.82 N (1 kg at Earth surface)
Ether inflow = escape speed	$v = \sqrt{(2GM/r)}$	$v = \sqrt{(c^5 M / (2\pi^2 \hbar v_e H r))}$	$v = \sqrt{(c^5 M / (2\pi^2 \hbar \cdot 8.71\times10^{84} \cdot r))}$	11,190 m/s (Earth surface)
Gravitational redshift	$\Delta f/f = GM/(rc^2)$	$\Delta f/f = f_M f_r / (v_e H)$, $f_M = Mc^2/h$, $f_r = c/2\pi r$	$\Delta f/f = f_M f_r / 8.71\times10^{84}$	6.961×10^{-10} (Earth surface)
Schwarzschild radius	$r_s = 2GM/c^2$	$r_s = c^3 M / (2\pi^2 \hbar v_e H)$	$r_s = c^3 M / (2\pi^2 \hbar \cdot 8.71\times10^{84})$	2954 m (Sun)
Planck length	$\ell_P = \sqrt{(\hbar G/c^3)}$	$\ell_P = c / (2\pi \sqrt{(v_e H)})$	$\ell_P = c / (2\pi \sqrt{(8.71\times10^{84})})$	1.616×10^{-35} m
Planck time	$t_P = \sqrt{(\hbar G/c^5)}$	$t_P = 1 / (2\pi \sqrt{(v_e H)})$	$t_P = 1 / (2\pi \sqrt{(8.71\times10^{84})})$	5.391×10^{-44} s
Planck mass	$m_P = \sqrt{(\hbar c/G)}$	$m_P = 2\pi \hbar \sqrt{(v_e H)} / c^2$	$m_P = 2\pi \hbar \sqrt{(8.71\times10^{84})} / c^2$	2.176×10^{-8} kg
Black-hole entropy	$S = k_B c^3 A / (4\hbar G)$	$S = \pi^2 k_B v_e H A / c^2$	$S = \pi^2 k_B \cdot 8.71\times10^{84} \cdot A / c^2$	1.321×10^{46} J/K (A = 1 m²)
Hawking temperature	$T = \hbar c^3 / (8\pi G M k_B)$	$T = \pi \hbar^2 v_e H / (2c^2 M k_B)$	$T = \pi \hbar^2 \cdot 8.71\times10^{84} / (2c^2 M k_B)$	6.168×10^{-8} K (solar-mass black hole)
Fine-structure constant	$\alpha = e^2/(4\pi\varepsilon_0\hbar c)$	$\alpha = \pi e^2 G v_e H / (\varepsilon_0 c^6)$	$\alpha = \pi e^2 G \cdot 8.71\times10^{84} / (\varepsilon_0 c^6)$	7.297×10^{-3} (dimensionless; e = electron charge)
Bohr radius	$a_0 = 4\pi\varepsilon_0\hbar^2 / (m_e e^2)$	$a_0 = \varepsilon_0 c^{10} / (4\pi^3 G^2 v_e^2 H^2 m_e e^2)$	$a_0 = \varepsilon_0 c^{10} / (4\pi^3 G^2 (8.71\times10^{84})^2 m_e e^2)$	5.292×10^{-11} m
Stefan–Boltzmann constant	$\sigma = \pi^2 k_B^4 / (60\hbar^3 c^2)$	$\sigma = \pi^2 k_B^4 (4\pi^2 G v_e H)^3 / (60 c^{17})$	$\sigma = \pi^2 k_B^4 (4\pi^2 G \cdot 8.71\times10^{84})^3 / (60 c^{17})$	5.670×10^{-8} W m^{-2} K^{-4}
Hubble law	$v = H d$	$v = H d$ (H = universe-rung frequency)	$v = 2.01\times10^{-18}$ s^{-1} · d	62,000 m/s (d = 1 Mpc)
Normalization (Zhang 2026) on Hubble sphere	$4\pi^2 r^3 c^2/(T^2 G h v) = 1$	$r = c/H$, $T = 2\pi/H \Rightarrow v = c^5/(hGH) = 2\pi v_e$	$v = 2\pi \cdot 4.34\times10^{102}$	2.725×10^{103} Hz
Temperature → frequency	$v = k_B T/h$	$v = 2\pi k_B T G v_e H / c^5$	$v = 2\pi k_B T G \cdot 8.71\times10^{84} / c^5$	6.459×10^{12} Hz (T = 310 K)
Two knots: gravity	$F = Gm_1m_2/r^2$	$F = c h f_1 f_2 / (2\pi v_e H r^2)$	$F = c h f_1 f_2 / (2\pi \cdot 8.71\times10^{84} \cdot r^2)$	6.674×10^{-11} N (two 1 kg, 1 m)
Electric / gravity ratio	$F_e/F_g = e^2/(4\pi\varepsilon_0 G m_1 m_2)$	$F_e/F_g = \alpha v_e H / (f_1 f_2)$	$F_e/F_g = \alpha \cdot 8.71\times10^{84} / (f_1 f_2)$	1.236×10^{36} (two protons)

Electric = gravity crossover	Stoney mass $\sqrt{\alpha}\, m_P$	$f = \sqrt{(\alpha\, v_e\, H)}$, $m = h f / c^2$	$f = \sqrt{(\alpha \cdot 8.71\times10^{84})}$	1.859×10^{-9} kg

Each ether form is an algebraic rewriting via the middle-rung law; the value column was checked to agree with the standard form to 10^{-9}. The physical content lies in the interpretation and in the dated predictions (P1–P5), not in the numerical agreement.

3. Chemistry formulas (natural constant e appears only in exponentials and logarithms)

Formula	Form
Ideal gas	$PV = nRT$, $R = N_A k_B$
Boltzmann distribution	$n_2/n_1 = e^{-h\Delta\nu/k_BT}$
Arrhenius	$k = A\, e^{-E_a/RT}$
Nernst	$E = E^0 - (RT/zF)\ln Q$; $2.303\, k_BT/e = 61.5$ mV per decade at 310 K
Gibbs / equilibrium	$\Delta G^0 = -RT \ln K$, $K = e^{-\Delta G^0/RT}$; $\Delta G = -nFE$
Proton-motive force	$\Delta p = \Delta\psi - (2.303RT/F)\,\Delta pH$
Radioactive decay	$N = N_0\, e^{-\lambda t}$, $t_{½} = \ln 2/\lambda$
Photon energy	$E = hc/\lambda = c^6 / (2\pi\, G\, v_e\, H\, \lambda)$
Rydberg constant	$R_\infty = \pi^3 m_e\, e^4\, G^3\, v_e^3\, H^3 / (\varepsilon_0^2\, c^{16})$
Bond energy → frequency	$\nu = \Delta E/h = 2\pi\, \Delta E\, G\, v_e\, H / c^5$

4. Elements as ether structures 1–118

Structure number = atomic number. Frequency $f = mc^2/h$. Ladder position $k = \log(m/m_U)/\log(m_P/m_U)$, with $m_U = hH/c^2$ and m_P the Planck mass. Ether form: $f = H^{1-k/2} \cdot v_e^{k/2}$ (exact). Values for $H_0 = 62.0$.

Z	Symbol	Atomic mass (u)	f (Hz)	k	Ether form
1	H	1.008	2.270×10^{23}	0.6823	$H^{0.6588}.v_e^{0.3412}$
2	He	4.0026	9.015×10^{23}	0.6923	$H^{0.6539}.v_e^{0.3461}$
3	Li	6.94	1.563×10^{24}	0.6962	$H^{0.6519}.v_e^{0.3481}$
4	Be	9.0122	2.030×10^{24}	0.6981	$H^{0.6509}.v_e^{0.3491}$
5	B	10.81	2.435×10^{24}	0.6994	$H^{0.6503}.v_e^{0.3497}$
6	C	12.011	2.705×10^{24}	0.7002	$H^{0.6499}.v_e^{0.3501}$
7	N	14.007	3.155×10^{24}	0.7013	$H^{0.6493}.v_e^{0.3507}$
8	O	15.999	3.604×10^{24}	0.7023	$H^{0.6489}.v_e^{0.3511}$
9	F	18.998	4.279×10^{24}	0.7035	$H^{0.6482}.v_e^{0.3518}$
10	Ne	20.18	4.545×10^{24}	0.7039	$H^{0.6480}.v_e^{0.3520}$

11	Na	22.99	5.178×10^{24}	0.7049	$H^{0.6476}.v_e^{0.3524}$
12	Mg	24.305	5.474×10^{24}	0.7053	$H^{0.6474}.v_e^{0.3526}$
13	Al	26.982	6.077×10^{24}	0.7060	$H^{0.6470}.v_e^{0.3530}$
14	Si	28.085	6.326×10^{24}	0.7063	$H^{0.6468}.v_e^{0.3532}$
15	P	30.974	6.976×10^{24}	0.7070	$H^{0.6465}.v_e^{0.3535}$
16	S	32.06	7.221×10^{24}	0.7073	$H^{0.6464}.v_e^{0.3536}$
17	Cl	35.45	7.985×10^{24}	0.7080	$H^{0.6460}.v_e^{0.3540}$
18	Ar	39.95	8.998×10^{24}	0.7089	$H^{0.6456}.v_e^{0.3544}$
19	K	39.098	8.806×10^{24}	0.7087	$H^{0.6456}.v_e^{0.3544}$
20	Ca	40.078	9.027×10^{24}	0.7089	$H^{0.6455}.v_e^{0.3545}$
21	Sc	44.956	1.013×10^{25}	0.7097	$H^{0.6451}.v_e^{0.3549}$
22	Ti	47.867	1.078×10^{25}	0.7102	$H^{0.6449}.v_e^{0.3551}$
23	V	50.942	1.147×10^{25}	0.7106	$H^{0.6447}.v_e^{0.3553}$
24	Cr	51.996	1.171×10^{25}	0.7108	$H^{0.6446}.v_e^{0.3554}$
25	Mn	54.938	1.237×10^{25}	0.7112	$H^{0.6444}.v_e^{0.3556}$
26	Fe	55.845	1.258×10^{25}	0.7113	$H^{0.6444}.v_e^{0.3556}$
27	Co	58.933	1.327×10^{25}	0.7117	$H^{0.6442}.v_e^{0.3558}$
28	Ni	58.693	1.322×10^{25}	0.7117	$H^{0.6442}.v_e^{0.3558}$
29	Cu	63.546	1.431×10^{25}	0.7122	$H^{0.6439}.v_e^{0.3561}$
30	Zn	65.38	1.473×10^{25}	0.7124	$H^{0.6438}.v_e^{0.3562}$
31	Ga	69.723	1.570×10^{25}	0.7129	$H^{0.6436}.v_e^{0.3564}$
32	Ge	72.63	1.636×10^{25}	0.7132	$H^{0.6434}.v_e^{0.3566}$
33	As	74.922	1.688×10^{25}	0.7134	$H^{0.6433}.v_e^{0.3567}$
34	Se	78.971	1.779×10^{25}	0.7138	$H^{0.6431}.v_e^{0.3569}$
35	Br	79.904	1.800×10^{25}	0.7139	$H^{0.6431}.v_e^{0.3569}$
36	Kr	83.798	1.887×10^{25}	0.7142	$H^{0.6429}.v_e^{0.3571}$
37	Rb	85.468	1.925×10^{25}	0.7144	$H^{0.6428}.v_e^{0.3572}$
38	Sr	87.62	1.974×10^{25}	0.7145	$H^{0.6427}.v_e^{0.3573}$
39	Y	88.906	2.002×10^{25}	0.7147	$H^{0.6427}.v_e^{0.3573}$
40	Zr	91.224	2.055×10^{25}	0.7148	$H^{0.6426}.v_e^{0.3574}$

41	Nb	92.906	2.093×10^{25}	0.7150	$H^{0.6425}.v_e^{0.3575}$
42	Mo	95.95	2.161×10^{25}	0.7152	$H^{0.6424}.v_e^{0.3576}$
43	Tc	98	2.207×10^{25}	0.7154	$H^{0.6423}.v_e^{0.3577}$
44	Ru	101.07	2.276×10^{25}	0.7156	$H^{0.6422}.v_e^{0.3578}$
45	Rh	102.91	2.318×10^{25}	0.7157	$H^{0.6421}.v_e^{0.3579}$
46	Pd	106.42	2.397×10^{25}	0.7159	$H^{0.6420}.v_e^{0.3580}$
47	Ag	107.87	2.430×10^{25}	0.7160	$H^{0.6420}.v_e^{0.3580}$
48	Cd	112.41	2.532×10^{25}	0.7163	$H^{0.6418}.v_e^{0.3582}$
49	In	114.82	2.586×10^{25}	0.7165	$H^{0.6418}.v_e^{0.3582}$
50	Sn	118.71	2.674×10^{25}	0.7167	$H^{0.6416}.v_e^{0.3584}$
51	Sb	121.76	2.742×10^{25}	0.7169	$H^{0.6415}.v_e^{0.3585}$
52	Te	127.6	2.874×10^{25}	0.7173	$H^{0.6414}.v_e^{0.3586}$
53	I	126.9	2.858×10^{25}	0.7172	$H^{0.6414}.v_e^{0.3586}$
54	Xe	131.29	2.957×10^{25}	0.7175	$H^{0.6413}.v_e^{0.3587}$
55	Cs	132.91	2.994×10^{25}	0.7176	$H^{0.6412}.v_e^{0.3588}$
56	Ba	137.33	3.093×10^{25}	0.7178	$H^{0.6411}.v_e^{0.3589}$
57	La	138.91	3.129×10^{25}	0.7179	$H^{0.6411}.v_e^{0.3589}$
58	Ce	140.12	3.156×10^{25}	0.7179	$H^{0.6410}.v_e^{0.3590}$
59	Pr	140.91	3.174×10^{25}	0.7180	$H^{0.6410}.v_e^{0.3590}$
60	Nd	144.24	3.249×10^{25}	0.7181	$H^{0.6409}.v_e^{0.3591}$
61	Pm	145	3.266×10^{25}	0.7182	$H^{0.6409}.v_e^{0.3591}$
62	Sm	150.36	3.387×10^{25}	0.7184	$H^{0.6408}.v_e^{0.3592}$
63	Eu	151.96	3.423×10^{25}	0.7185	$H^{0.6407}.v_e^{0.3593}$
64	Gd	157.25	3.542×10^{25}	0.7188	$H^{0.6406}.v_e^{0.3594}$
65	Tb	158.93	3.580×10^{25}	0.7188	$H^{0.6406}.v_e^{0.3594}$
66	Dy	162.5	3.660×10^{25}	0.7190	$H^{0.6405}.v_e^{0.3595}$
67	Ho	164.93	3.715×10^{25}	0.7191	$H^{0.6404}.v_e^{0.3596}$
68	Er	167.26	3.767×10^{25}	0.7192	$H^{0.6404}.v_e^{0.3596}$
69	Tm	168.93	3.805×10^{25}	0.7193	$H^{0.6404}.v_e^{0.3596}$
70	Yb	173.05	3.898×10^{25}	0.7195	$H^{0.6403}.v_e^{0.3597}$

71	Lu	174.97	3.941×10^{25}	0.7195	$H^{0.6402}.\nu_e{}^{0.3598}$
72	Hf	178.49	4.020×10^{25}	0.7197	$H^{0.6402}.\nu_e{}^{0.3598}$
73	Ta	180.95	4.076×10^{25}	0.7198	$H^{0.6401}.\nu_e{}^{0.3599}$
74	W	183.84	4.141×10^{25}	0.7199	$H^{0.6401}.\nu_e{}^{0.3599}$
75	Re	186.21	4.194×10^{25}	0.7200	$H^{0.6400}.\nu_e{}^{0.3600}$
76	Os	190.23	4.285×10^{25}	0.7201	$H^{0.6399}.\nu_e{}^{0.3601}$
77	Ir	192.22	4.329×10^{25}	0.7202	$H^{0.6399}.\nu_e{}^{0.3601}$
78	Pt	195.08	4.394×10^{25}	0.7203	$H^{0.6398}.\nu_e{}^{0.3602}$
79	Au	196.97	4.436×10^{25}	0.7204	$H^{0.6398}.\nu_e{}^{0.3602}$
80	Hg	200.59	4.518×10^{25}	0.7205	$H^{0.6397}.\nu_e{}^{0.3603}$
81	Tl	204.38	4.603×10^{25}	0.7207	$H^{0.6397}.\nu_e{}^{0.3603}$
82	Pb	207.2	4.667×10^{25}	0.7208	$H^{0.6396}.\nu_e{}^{0.3604}$
83	Bi	208.98	4.707×10^{25}	0.7208	$H^{0.6396}.\nu_e{}^{0.3604}$
84	Po	209	4.707×10^{25}	0.7208	$H^{0.6396}.\nu_e{}^{0.3604}$
85	At	210	4.730×10^{25}	0.7209	$H^{0.6396}.\nu_e{}^{0.3604}$
86	Rn	222	5.000×10^{25}	0.7213	$H^{0.6394}.\nu_e{}^{0.3606}$
87	Fr	223	5.023×10^{25}	0.7213	$H^{0.6394}.\nu_e{}^{0.3606}$
88	Ra	226	5.090×10^{25}	0.7214	$H^{0.6393}.\nu_e{}^{0.3607}$
89	Ac	227	5.113×10^{25}	0.7214	$H^{0.6393}.\nu_e{}^{0.3607}$
90	Th	232.04	5.226×10^{25}	0.7216	$H^{0.6392}.\nu_e{}^{0.3608}$
91	Pa	231.04	5.204×10^{25}	0.7215	$H^{0.6392}.\nu_e{}^{0.3608}$
92	U	238.03	5.361×10^{25}	0.7218	$H^{0.6391}.\nu_e{}^{0.3609}$
93	Np	237	5.338×10^{25}	0.7217	$H^{0.6391}.\nu_e{}^{0.3609}$
94	Pu	244	5.496×10^{25}	0.7219	$H^{0.6390}.\nu_e{}^{0.3610}$
95	Am	243	5.473×10^{25}	0.7219	$H^{0.6390}.\nu_e{}^{0.3610}$
96	Cm	247	5.563×10^{25}	0.7220	$H^{0.6390}.\nu_e{}^{0.3610}$
97	Bk	247	5.563×10^{25}	0.7220	$H^{0.6390}.\nu_e{}^{0.3610}$
98	Cf	251	5.653×10^{25}	0.7221	$H^{0.6389}.\nu_e{}^{0.3611}$
99	Es	252	5.676×10^{25}	0.7222	$H^{0.6389}.\nu_e{}^{0.3611}$
100	Fm	257	5.789×10^{25}	0.7223	$H^{0.6388}.\nu_e{}^{0.3612}$

101	Md	258	5.811×10^{25}	0.7223	$H^{0.6388}\cdot v_e^{0.3612}$
102	No	259	5.834×10^{25}	0.7224	$H^{0.6388}\cdot v_e^{0.3612}$
103	Lr	266	5.991×10^{25}	0.7226	$H^{0.6387}\cdot v_e^{0.3613}$
104	Rf	267	6.014×10^{25}	0.7226	$H^{0.6387}\cdot v_e^{0.3613}$
105	Db	268	6.036×10^{25}	0.7226	$H^{0.6387}\cdot v_e^{0.3613}$
106	Sg	269	6.059×10^{25}	0.7226	$H^{0.6387}\cdot v_e^{0.3613}$
107	Bh	270	6.081×10^{25}	0.7227	$H^{0.6387}\cdot v_e^{0.3613}$
108	Hs	269	6.059×10^{25}	0.7226	$H^{0.6387}\cdot v_e^{0.3613}$
109	Mt	278	6.262×10^{25}	0.7229	$H^{0.6386}\cdot v_e^{0.3614}$
110	Ds	281	6.329×10^{25}	0.7230	$H^{0.6385}\cdot v_e^{0.3615}$
111	Rg	282	6.352×10^{25}	0.7230	$H^{0.6385}\cdot v_e^{0.3615}$
112	Cn	285	6.419×10^{25}	0.7231	$H^{0.6385}\cdot v_e^{0.3615}$
113	Nh	286	6.442×10^{25}	0.7231	$H^{0.6385}\cdot v_e^{0.3615}$
114	Fl	289	6.509×10^{25}	0.7232	$H^{0.6384}\cdot v_e^{0.3616}$
115	Mc	290	6.532×10^{25}	0.7232	$H^{0.6384}\cdot v_e^{0.3616}$
116	Lv	293	6.599×10^{25}	0.7233	$H^{0.6384}\cdot v_e^{0.3616}$
117	Ts	294	6.622×10^{25}	0.7233	$H^{0.6384}\cdot v_e^{0.3616}$
118	Og	294	6.622×10^{25}	0.7233	$H^{0.6384}\cdot v_e^{0.3616}$

5. Status

Predictions P1–P5 and the disclosure of data already examined are stated in the preregistration text. This compendium adds no new prediction; it records the formulas and present values as of 2026-11-05.